

EECS 182 Type E Participation Log

Assignment Option: E (Make AI-enhanced learning tools)

Tool Used: Google NotebookLM

Source Material: Prince, *Understanding Deep Learning*, Chapter 12: Transformers

Part 1: Executive Summary

Goal: To deepen my conceptual understanding of the mathematical intuitions behind Transformers (specifically Self-Attention and Positional Encodings) before the lecture.

Method: I created a custom "Socratic Tutor" using Google NotebookLM. Instead of asking for simple summaries, I prompted the AI to quiz me on specific design choices in the architecture (e.g., *why* scale the dot product? *why* add positional encodings instead of concatenating?).

Observations:

- **One-shot Success:** The AI successfully adopted the persona of a strict TA. [cite_start]It correctly identified that the "keys" and "queries" terminology originates from information retrieval [cite: 455] and explained the gradient saturation issue in Softmax without hallucinating.
- **Insight:** The interaction clarified the "subspace" intuition for positional encodings. [cite_start]This detail was hidden in the chapter notes [cite: 1304-1306], which I likely would have missed in a casual reading of the main text.
- **Verification:** The AI's responses were highly accurate, strictly referencing specific equations (Eq 12.5, 12.9) and figures (Fig 12.3) from the text.

Part 2: Annotated Log of Interaction

Student (Me):

You are a strict Teaching Assistant for EECS 182. Based on the uploaded Chapter 12 PDF, quiz me on the **Self-Attention Mechanism**. Specifically, ask me why we need three distinct vectors

(Queries q , Keys k , Values v) instead of just using the input vector x for everything. Don't give me the answer; check my intuition.

AI (NotebookLM):

Let's test your intuition on Section 12.2.

In Equation 12.5, we compute attention weights using $k^T q$. [cite_start]The text mentions that these names ("queries" and "keys") are inherited from the field of **Information Retrieval**[cite: 455].

Question: If we just used the input x for everything (i.e., $q = k = v = x$), what would that imply about the relationship between "searching" for information and "retrieving" content? [cite_start]Why does the text describe the Softmax function as a "competition"[cite: 456]?

Student (Me):

If we used x for everything, the attention would just be based on the dot product of the input with itself, so words would only attend to identical words. The Q, K, V separation allows us to route information differently.

AI (NotebookLM):

That's a good start, but let me refine your "identical words" point.

You are correct that Q and K allow for a distinct routing mechanism. [cite_start]As Figure 12.1 illustrates, self-attention is fundamentally about **routing** values[cite: 352]. [cite_start]The text specifies that the query q_n compares itself to all keys k_m to determine the similarity[cite: 455].

Critique: You missed a subtle point. Even if $q = k = x$, the dot product wouldn't *just* be high for identical words; it would be high for any vectors with large dot products (alignment). However, the separation allows the n -th output to "pay attention" to the m -th input based on a learned compatibility, separate from the content (v) being retrieved. The Softmax ensures these weights sum to one, creating the "competition".

Annotation: [Observation: The AI correctly distinguished between the *mechanism* of dot products and the *semantic role* of Q/K/V as described in Sec 12.2. It correctly cited the "routing" analogy from the text.]

Student (Me):

Okay, let's move to **Scaled Dot Product Attention** (Section 12.3.2). Quiz me on the scaling factor.

AI (NotebookLM):

Look at Equation 12.9. [cite_start]We divide the dot product $K^T Q$ by $\sqrt{D_q}$ [cite: 563].

Question: What specifically happens to the **Softmax** function if we *don't* do this scaling, especially when the dimension D_q is large? The text mentions a specific problem regarding training.

Student (Me):

(Intentionally answering vaguely) We divide by the square root to normalize the variance to 1, otherwise the numbers get too big and the matrix multiplication is unstable.

AI (NotebookLM):

You are half-right about the "numbers getting too big," but "matrix multiplication instability" is not the reason given in the text.

[cite_start]**Correction:** The text explicitly states that large magnitude dot products move the arguments of the softmax function into a region where "the largest value completely dominates" [cite: 560].

The Consequence: When this happens, small changes to the inputs have little effect on the output. [cite_start]This means the **gradients are very small**, making the model "difficult to train" [cite: 561]. It's a vanishing gradient problem, not just a numerical stability issue.

Annotation: [Observation: The AI successfully caught my vague answer. It pinpointed the specific consequence mentioned in the text: small gradients leading to difficult training, rather than just generic "instability".]

Student (Me):

Final question. In Section 12.3.1 regarding **Positional Encodings**, why do we **add** the position matrix Π to X ? Why don't we concatenate it? That seems destructive to the data.

AI (NotebookLM):

This is a common point of confusion.

Question: The notes section at the end of the chapter provides a geometric justification for this "addition". It argues that since the data dimension D is usually much larger than the sequence length N , the positional encoding lies in a specific type of geometric space relative to the word embeddings. [cite_start]What is that term? [cite: 1305]

Student (Me):

Is it because they lie in orthogonal subspaces?

AI (NotebookLM):

Exactly.

[cite_start]The notes explain that because $D > N$, the positional encoding lies in a **subspace**[cite: 1305]. [cite_start]Since the word embeddings X are learned, the model can theoretically learn to keep the semantic information and the positional information in **orthogonal subspaces**[cite: 1306], allowing it to retrieve both without interference. Concatenation isn't strictly necessary because the high-dimensional space has enough "room" for both.

Annotation: [Observation: Impressive behavior. The AI retrieved information from the "Notes" section (Page 236) to answer this advanced question, demonstrating it indexed the entire PDF, not just the main body.]